Gagana M D

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Early-career AI safety researcher working on monitoring and evaluating AI agents: does a monitor's evidence arrive in time to act on, and does an evaluation support the claim made from it? I build controlled experiments and the code behind them.

Research Experience

PRISM AI Safety Research Fellowship – *Research Fellow, Remote* May 2026 – Present

- **LLM-generated world models** (mentor: Jobst Heitzig). Main code contributor (78 of ∼116 commits) to a 44-scenario benchmark of five LLMs writing executable multi-agent games (POSGs). 134 of 135 games passed mechanical checks, yet mean structural F1 against reference games was 0.37. Paper under review at a NeurIPS 2026 workshop.
- Research support for two fellowship teams (interpretability; AI control): experiment tracking, reviews, milestones.

Sentient Futures – *Research Fellow, Remote (mentor: Julia Bossmann)* Sep 2026 – Present

- **Gap Attractor Project.** Designing behavioural evaluations of whether language models show stable, self-directed curiosity (unprompted information-seeking and question generation) that persists across prompts and controls.
- **Post-Control Initiative.** Lead interdisciplinary roundtables (AI safety, governance, game theory) on human–AI cooperation when reliable control can't be assumed, and write the resulting research syntheses.

FlowResearch – *Research Fellow, Remote* Apr 2026 – Present

- Behavioural evaluation of RL and world-model-based agents: planning, generalisation and safety-relevant behaviour.

BaseThesis Labs – *AI Safety Researcher, Evaluations and Benchmarking, Bengaluru* May 2026 – Jul 2026

- Benchmarked AI agents on long-horizon and ultra-long-horizon CFO and financial due-diligence tasks. Built the offline-first harness: rubric grading with safety dealbreakers, a ground-truth leakage guard, trace capture, state-diff auditing, and 50 tests run in CI, so every score can be traced back to what the agent actually did. Code.
- Measured when agent orchestration pays off across coding, ledger reconciliation and a 717-item diligence corpus. It cost 3.5× the latency and was worth it only where coverage and traceability mattered. Code.

Indian Institute of Science, DREAM Lab – *Research Intern, Bengaluru* Jan 2025 – Aug 2025

- Reinforcement-learning policies that split model inference across edge devices (ExpansionNet v2, YOLOv5), with device profiling and latency-aware scheduling. Distributed optimisation improved system utility by 31%.

Neuromatch – *Impact Scholar, Remote* Oct 2024 – Mar 2025

- Co-authored a study of whether transformer-based RL agents follow the Matching Law (reward ratios 1:1 to 16:1).

Selected Project

Detection Is Not Containment – Apart Research AI Incident Response Sprint Sep 2026

- First author; built the monitoring experiment. Does a monitor flag a prohibited agent action early enough for revoking authority to prevent it? Hash-frozen oracle and prompt, benign-only threshold calibration, preregistered phases.
- Detection preceded the last safe point by one step; meaning-preserving renames kept only 8/12 decisions; a replayed false alarm failed a benign task. Case study, code.

Publications and Reports

Gagana M D, Ajay Kumar K, Vijayashekhar S S, Jignesh S. Bhatt. *Hyperspectral BiorNet: A Complete Wavelet Framework for Unmixing.* IEEE InGARSS 2025, pp. 15–19. DOI

Gagana M D, Kaustubh Gupta, María Inés Berdiñas. *Detection Is Not Containment: Timely Monitoring Determines Whether AI Agent Failures Can Be Prevented.* Apart Research, 2026. Report

Y. Karakoc, E. Sharifi, **M. D. Gagana**, Z. Sarayloo. *Investigating the Matching Law in Transformer-Based Reinforcement Learning Agents.* Zenodo, 2025. DOI

Paper on the structural fidelity of LLM-generated world models. Under review, NeurIPS 2026 workshop.

Education

B.E., Artificial Intelligence and Machine Learning – Acharya Institute of Technology, CGPA 8.38/10 2021 – 2025

Also: Neuromatch Academy, Computational Neuroscience (2025) and Deep Learning (2024).

Honours, Skills and Community

Honours: 2nd place, AMD AI Sprint Hackathon (AMD, Meta, Hugging Face), adversarial Q&A agents; 1st place, Federated Learning Client Selection Challenge, IISc; 1st place, Srushti 2025 state-level project exhibition.

Tools: Python, PyTorch, Transformers, Stable-Baselines3, Ollama, pytest, GitHub Actions, Docker, FastAPI, Git, Linux.

Community: Women in Big Data India, Shakti Mentorship Program team and workshop facilitator (2026–); WiDS Ambassador (2026–); Women Who Code Data, helped run 11+ workshops reaching 350+ people (2022–24).